
Conditionally Selected Platform Configuration and Deployment Baseline

This document defines the conditionally selected fixed-site spectrum-sensing configuration based on the HackRF One R10 and Raspberry Pi 5 for *Template 1S: Fixed Single-Node Monitoring*. Named products identify the selected baseline implementation; controlled substitutions may be approved only when they satisfy the same functional, interface, and acceptance requirements.

Baseline status: Conditionally selected. The SDR and compute platforms are selected. The complete operational-node baseline remains conditional on closure of the sensing-performance criteria, scan plan, RF front end, timing interface, secure network configuration, software and cybersecurity controls, site installation, controlled bill of materials, cost baseline, calibration, and applicable acceptance evidence.

Deployment Requirements

The approved monitoring plan, MP-01, establishes the operational requirements for *Template 1S: Fixed Single-Node Monitoring*. The selected hardware and software configuration defined later in this document shall satisfy these requirements without expanding the approved scope.

- MP-01-PUR *Monitoring purpose and intended use.*

The node shall operate continuously at one fixed site and shall perform sequential scheduled screening of the approved tuning list. Continuous node operation shall not be interpreted as simultaneous or uninterrupted observation of every approved band.

The node shall generate screening-grade records for identifying candidate events and supporting preliminary technical review. Node records shall not independently establish regulatory non-compliance. Formal compliance conclusions shall require an approved calibrated measurement and evidence-assurance procedure.

- MP-01-JUR *Jurisdiction and responsible organization.*

The node shall operate within the applicable Colombian spectrum-management framework and under the responsibility of the organization that owns or operates the monitoring system. No authorization, delegation, endorsement, or operational responsibility by the Agencia Nacional del Espectro (ANE) shall be inferred unless documented in a controlled authorization record.

- MP-01-BND *Monitored services, bands, and exclusions.*

The authoritative monitoring-band list is defined in Appendix A. Exact channels, tuning positions, priorities, and effective assignments shall be controlled through SP-01.

The 608–614 MHz interval shall remain excluded from the approved DTT scan range. The 5350–5470 MHz interval shall remain excluded from the approved 5-GHz WAS/RLAN scan range unless a separately authorized monitoring requirement is established.

- **MP-01-SCN** *Scan coverage and observation schedule.*

The controlled scan plan, SP-01, shall define, for every monitored service, the tuning positions, channel bandwidths, sample rates, dwell times, revisit intervals, retuning and settling allowances, processing intervals, maximum scan-cycle duration, exclusion intervals, scan priorities, event-interruption behavior, and recovery rules.

SP-01 shall also state the minimum event duration expected to be observable and the accepted risk of missing short-duration events created by sequential scanning. It shall be derived from MP-01, registered in VTM-01, and verified as feasible before operational acquisition begins.

Portable survey operation, simultaneous multiband observation, distributed synchronization, direction finding, TDOA, and emitter geolocation are outside the approved baseline.

- **MP-01-MET** *Required measurands and derived records.*

The node shall generate received-power estimates, center-frequency estimates, occupied-bandwidth estimates, spectral-centroid estimates, signal-duration records, duty-cycle estimates, channel-occupancy statistics, noise-floor estimates, signal-to-noise-ratio estimates, host event timestamps, power spectral densities, spectrograms, and event-summary records.

BA-01-MET shall define the applicable accuracy, uncertainty, detection probability, false-alarm rate, event-detection latency, normalization, repeatability, and decision criteria for these outputs.

- **MP-01-ACQ** *Acquisition and continuity requirements.*

The acquisition system shall provide complex-IQ data at a sample rate and effective bandwidth sufficient for the measurands assigned to each tuning position. The selected continuous baseline mode is defined in Table 2.

Alternate or burst modes shall not be used operationally until sustained transfer, processing, storage throughput, sample continuity, timing, and thermal stability have been verified under BA-01-ACQ.

The platform shall detect, timestamp, and record sample discontinuities, receiver overload, clipping, invalid calibration states, storage congestion, and processing-queue overflow.

- **MP-01-TIM** *Timing, timestamping, and location.*

The 10 MHz reference shall support receiver-frequency stability. The GNSS-referenced 1 PPS signal and approved host timing interface shall support UTC-referenced host event timestamping.

Event records shall provide a timestamp resolution of 1 ms or better. BA-01-TIM shall separately define the maximum timestamp error, timestamp origin, processing-latency treatment, holdover behavior, discontinuity handling, and reported uncertainty. The baseline shall not claim UTC-referenced sample timestamps.

The verified coordinates of the fixed site and the timing-source, reference-lock, timestamp-validity, and discontinuity states shall be recorded with each applicable observation.

- **MP-01-ENV** *Operating environment and installation.*

The node shall operate in dense urban RF conditions containing strong broadcast and cellular signals, weak adjacent-channel emissions, impulsive interference, multipath propagation, intermittent transmitters, near-far conditions, and temperature-dependent receiver variation.

The equipment shall be installed in a protected fixed-site enclosure that permits safe inspection, maintenance, and replacement. The complete node shall operate within the environmental limits defined in BA-01-ENV.

- **MP-01-CTX** *Contextual-data requirements.*

Contextual data shall include authorized frequency assignments, transmitter-registration records, service-allocation tables, site coordinates, antenna records, observation schedules, equipment inventories, and applicable technical-limit documents.

Every source shall identify its issuing authority, document identifier, revision, effective date, geographic scope, retrieval date, applicability, and freshness status.

- **MP-01-DAT** *Processing, storage, communication, and retention.*

The node shall generate FFT, PSD, spectrogram, occupancy, health, and event-summary products locally. Raw IQ shall be retained only following an approved event trigger, scheduled validation activity, or authorized operator request.

The routine-retention period for metadata, PSD, spectrogram, occupancy, health, audit, and event-summary records shall be 90 days. The controlled retention policy shall define raw-IQ limits, quotas, overwrite rules, storage-reserve thresholds, storage-exhaustion behavior, and investigation record retention.

Network transfer shall be restricted to authorized metadata, derived products, health records, events, and selected-IQ records.

- **MP-01-PWR** *Power and thermal constraints.*

Measured complete-node input power shall remain below 50 W under the maximum approved workload. The processing host shall use active cooling, and the power and thermal configuration shall satisfy BA-01-PWR and BA-01-ENV.

- **MP-01-SEC** *Cybersecurity, access control, and audit.*

The node shall implement role-based access control, unique operator accounts, key-based remote login, authenticated and encrypted administrative and data-transfer connections, controlled software updates, versioned configuration changes, restricted service ports, backup and restoration, and unauthorized-change detection.

Audit records shall include authentication attempts, administrative actions, configuration changes, software updates, service-state changes, and data exports.

- **MP-01-EVD** *Calibration, provenance, and evidence assurance.*

The node shall record equipment identifiers, RF-path configuration, antenna and cable identifiers, gain and attenuation states, calibration identifiers and validity periods, software and firmware versions, sample-continuity indicators, processing history, timing validity, record hashes, authorized modifications, and export history.

Template 5 shall apply only to records formally escalated under MP-01. Such records shall follow the approved chain-of-custody, integrity, uncertainty, retention, access-control, and controlled-export requirements.

- **MP-01-CST** *Cost and lifecycle basis.*

The USD 1,000–2,000 target shall apply to recurring procurement and node-specific integration cost per operational node. It shall include controlled hardware, assembly, node-specific commissioning, shipping, importation, taxes, initial spares, and contingency. It shall exclude separately controlled non-recurring engineering, shared laboratory qualification, and five-year lifecycle support.

If the closed recurring per-node cost exceeds USD 2,000, the project shall document an approved budget exception, revise only optional capabilities, or reopen the platform-selection decision. Mandatory safety, protection, acceptance, calibration, cybersecurity, and evidence controls shall not be removed solely to satisfy the target.

The five-year lifecycle budget shall separately include calibration, replacement storage, software maintenance, spare units, field support, site maintenance, backup resources, and component-obsolescence provisions.

Phase-0 Controlled-Artifact Registration and Traceability

BA-01 is the controlled *Baseline Acceptance Specification* for the conditionally selected platform configuration. It shall be approved, placed under configuration control, registered in the Phase-0 controlled-artifact registry, and anchored in VTM-01 before Stage 3 acceptance testing begins.

The Stage 1 feasibility record, FRM-01, and the Stage 2 comparative ranking record, CRM-01, shall be completed and registered in VTM-01 before Stage 3.

The controlled acceptance sequence shall be:

$$\begin{aligned} \text{MP-01} &\longrightarrow \{\text{SP-01}, \text{BA-01}\} \\ &\longrightarrow \text{Q3-01} \longrightarrow \text{T0-*} \longrightarrow \text{VTM-01} \longrightarrow \text{EAM-01}. \end{aligned}$$

Q3-01 shall qualify the laboratory environment used for calibration, RF-path characterization, and acceptance testing. A5-01 shall apply only when Template 5 evidence assurance is invoked for formally escalated records.

BA-01 Minimum Controlled Content BA-01 shall identify:

- the accepted configuration items and revisions;
- the applicable MP-01 and SP-01 identifiers;
- the corresponding acceptance criteria and T0-* procedures;
- the mandatory sensing cases, RF paths, gain and attenuation states, acquisition modes, storage configurations, timing states, workloads, security states, site conditions, and environmental conditions;
- the quantitative thresholds, measurement uncertainty, guard bands, and decision rules;
- the required calibration, configuration, provenance, integrity, cybersecurity, software, site, and audit evidence;
- the treatment of invalid, incomplete, repeated, conditional, and superseded tests;
- the responsible test, review, and approval authorities; and
- the evidence package required for the final EAM-01 decision.

Required VTM-01 Anchors The identifiers in Table 1 shall be used as controlled BA-01 section identifiers and as the corresponding VTM-01 traceability anchors.

Each anchor shall be dispositioned in EAM-01 as *Accepted*, *Conditionally Accepted*, or *Rejected*, supported by the applicable controlled evidence.

Any change to BA-01, SP-01, or the selected configuration after test execution begins shall require a documented impact assessment, versioned approval, and identification of all tests and evidence that must be repeated, revalidated, or superseded.

Baseline Hardware Infrastructure

The controlled hardware baseline shall comprise the configuration in Table 2. Device capabilities that exceed the approved operational configuration shall not be interpreted as accepted operating modes.

Configuration Control and Acceptance The controlled procurement bill of materials shall identify the manufacturer, model, part number, revision, electrical and environmental ratings, connector or interface type, mounting method, and operating-temperature range for every baseline component.

The Raspberry Pi 5 implementation shall use bounded processing queues, sample-continuity monitoring, controlled buffering, and health monitoring. Acceptance shall be performed under simultaneous acquisition, processing, storage, network, timing, cooling, and RF-control operation.

No hardware item, interface, gain state, acquisition mode, or environmental configuration shall be released for operational use until the applicable BA-01-ACQ, BA-01-RF, BA-01-TIM, BA-01-STR, BA-01-PWR, BA-01-ENV, BA-01-SW, BA-01-SEC, and BA-01-SIT criteria have been satisfied.

Selected RF Front End

The controlled RF chain shall be:

Antenna → Primary antenna-line surge protection → Antenna selector
→ Secondary limiter/DC block → Preselector → LNA/Bypass
→ Attenuator/Bypass → HackRF One.

The RF front end shall protect the receiver, limit out-of-band loading, and provide reproducible RF-path, gain, and attenuation states. Every selected state shall be recorded with the corresponding monitoring record.

Requirement Group	BA-01 Anchor	Required Verification Evidence
Sensing and measurands	BA-01-MET	Detection, false-alarm, latency, power, frequency, bandwidth, noise-floor, SNR, occupancy, PSD, repeatability, and uncertainty tests
Acquisition and continuity	BA-01-ACQ	Sample-continuity, sustained-throughput, scan-cycle, dwell, retuning, queue, and maximum-workload tests
RF-path performance	BA-01-RF	Insertion-loss, rejection, isolation, return-loss, overload, sensitivity, gain-state, attenuation-state, switching, and calibration tests
Timing and timestamping	BA-01-TIM	Frequency-reference, event-timestamp resolution and error, reference-lock, holdover, reacquisition, and continuity tests
Storage and records	BA-01-STR	Throughput, retention, quota, congestion, interrupted-write, recovery, and record-integrity tests
Power and fault recovery	BA-01-PWR	Load, regulation, ripple, startup, interruption, branch-protection, isolation, and recovery tests
Environment and enclosure	BA-01-ENV	Thermal, humidity, cooling-failure, enclosure, grounding, bonding, surge-protection, and environmental tests
Software and services	BA-01-SW	Configuration reproducibility, processing correctness, bounded-queue, service-startup, restart, watchdog, degraded-state, and recovery tests
Cybersecurity and audit	BA-01-SEC	Authentication, authorization, encryption, restricted-service, update, backup, restoration, audit, export-control, and unauthorized-change tests
Site commissioning	BA-01-SIT	Installed antenna, cable, grounding, bonding, surge-protection, timing, network, enclosure, cooling, access-control, and site tests
Evidence assurance	BA-01-EVD	Provenance, calibration validity, integrity, chain-of-custody, retention, access-control, and controlled-export checks, including A5-01 where Template 5 is invoked

Table 1: Minimum VTM-01 anchors for BA-01-controlled acceptance

Controlled Item	Baseline Selection	Controlled Implication
SDR platform	HackRF One R10; 1 MHz–6 GHz device range; 8-bit ADC; USB 2.0; half-duplex; 10 MHz CLKIN	Approved for receive-only monitoring. Every RF path and gain state shall be characterized under BA-01-RF . USB continuity and overload behavior shall be verified under BA-01-ACQ
Continuous acquisition	Single-channel complex IQ at 10 Msps using signed 8-bit interleaved samples	This is the approved continuous mode. Higher sample rates are alternate or burst modes and require separate BA-01-ACQ verification
Compute platform	Raspberry Pi 5 with 8 GB RAM and active cooling	Shall support bounded buffering, FFT/PSD processing, detection, metadata generation, storage, and network operation under the maximum approved simultaneous workload
Local storage	At least 2 TB of approved NVMe-class storage connected through an approved interface	Sustained throughput, congestion handling, retention, recovery, and record integrity shall satisfy BA-01-STR
Timing reference	GNSS-disciplined 10 MHz frequency reference and 1 PPS timing signal	Frequency stability, reference-lock status, timestamp error, and timing continuity shall satisfy BA-01-TIM
Network interface	Gigabit Ethernet through a protected enclosure entry	Supports authenticated and encrypted transfer of authorized metadata, derived records, events, health records, and selected IQ
Control interfaces	Raspberry Pi 5 GPIO, SPI, and I ² C interfaces	Supports antenna and preselector switching, attenuation and LNA control, timing/status inputs, and subsystem health monitoring
Power architecture	Regulated complete-node source rated for at least 40 W continuous output, including a dedicated 5 V/5 A Raspberry Pi rail and independently protected auxiliary outputs	Startup current, transient response, voltage regulation, branch protection, interruption recovery, and maximum-load operation shall satisfy BA-01-PWR
Installation and environmental protection	Lockable fixed-site enclosure with active cooling, temperature monitoring, protected RF, power, timing, and	Thermal performance, environmental protection, grounding, bonding, surge-current routing, cable support, and safe

Antenna	Frequency Range	Controlled Application
BicoLOG 5070	70–700 MHz	Approved VHF, lower-UHF, land-mobile, and DTT monitoring paths
HyperLOG 7060	700 MHz–6 GHz	Approved 700–960 MHz, 2.4-GHz, and 5-GHz monitoring paths

Table 3: Controlled antenna configuration

Antenna Subsystem Each rooftop antenna feed shall include approved primary antenna-line surge protection before the antenna selector unless a controlled installation study demonstrates equivalent protection. An absorptive SPDT antenna selector or equivalent shall connect exactly one protected antenna feed to the downstream RF chain. Its insertion loss, isolation, switching time, power handling, fail-safe state, and environmental rating shall be verified under **BA-01-RF** and **BA-01-SIT**.

Protection and Preselector Assembly The antenna-selector output shall pass through the approved secondary limiter and DC block before entering the preselector. The preselector shall use paired absorptive SP8T input and output selectors operated synchronously. The eight selector positions shall comprise seven filtered paths and one protected bypass path.

The DTT and 5-GHz dual-passband assemblies are conditionally selected pending RF-feasibility closure. Each shall present one controlled two-port path between the paired SP8T selectors. Before configuration freeze, the project shall identify a realizable circuit or commercial implementation and shall close its passband, ripple, return-loss, rejection, isolation, power-handling, temperature-response, physical-integration, and procurement requirements. Successful prototype verification under **BA-01-RF** is mandatory.

The selector fail-safe state shall disconnect the HackRF input or select the protected bypass path. The Raspberry Pi 5 shall control the antenna and preselector selectors through an approved GPIO, SPI, or I²C interface.

Gain and Attenuation Assembly The selected preselector output shall pass through an approved LNA/bypass stage and an approved attenuator/bypass stage. **BA-01-RF** shall define the permitted state for every preselector position.

Every LNA, bypass device, and attenuator shall be identified in the controlled bill of materials and validated for frequency response, gain or attenuation accuracy, noise figure where applicable, compression, overload behavior, isolation, switching transients, settling time, temperature response, and cal-

Selector	Position	Controlled RF Path	Required Function
1		88–108 MHz FM-bandpass assembly	Support FM monitoring. Any notch substate shall be separately controlled, calibrated, and recorded
2		108–137 MHz aeronautical bandpass assembly	Support the approved aeronautical radionavigation and communications monitoring scope
3		406–470 MHz land-mobile bandpass assembly	Support the approved private land-mobile monitoring range
4		Two-port dual-passband DTT assembly for 470–608 MHz and 614–698 MHz	Support DVB-T2 channels 14–36 and 38–51 while rejecting 608–614 MHz
5		700–960 MHz UHF/cellular bandpass assembly	Support the approved 700-, 800-, and 900-MHz monitoring ranges
6		2400–2483.5 MHz ISM/WAS/RLAN bandpass assembly	Support the approved 2.4-GHz monitoring range
7		Two-port dual-passband 5-GHz WAS/RLAN assembly for 5150–5350 MHz and 5470–5850 MHz	Support the approved 5-GHz subbands while rejecting 5350–5470 MHz
8		Protected 50 Ω bypass path	Support approved maintenance, characterization, and bypass operation without removing receiver-input protection

Table 4: Controlled preselector positions

Component or State	Validated Range	Controlled Function
Passive LNA bypass	All approved bands	Default state where the accepted sensitivity criterion is satisfied without external amplification
Band-specific VHF LNA	88–137 MHz	Conditional weak-signal amplification for approved VHF paths
Band-specific lower-UHF LNA	406–500 MHz	Conditional amplification for the land-mobile path and lower portion of the DTT path
ZX60-33LN-S+ or equivalent	0.5–4 GHz	Conditional amplification only within its validated range
Band-specific 5-GHz LNA	Approved 5-GHz subbands	Conditional weak-signal amplification for the 5-GHz path
Attenuator bypass	All approved bands	Approved state where external attenuation is unnecessary and overload criteria remain satisfied
ZX76-31R75PP-S+ or equivalent	0.1–4 GHz	Programmable attenuation for approved sub-4-GHz paths
Band-specific 5-GHz attenuator	Approved 5-GHz subbands	Conditional attenuation where required by the accepted 5-GHz overload-control strategy

Table 5: Controlled gain, bypass, and attenuation states

ibration stability.

Gain-Control Strategy Gain-state thresholds, hysteresis, and minimum state-hold times shall be defined and verified under **BA-01-RF**. The implementation shall:

1. monitor FFT peak level, broadband RMS level, clipping, and overload indicators;
2. disable the LNA, select LNA bypass, increase attenuation, or select attenuator bypass according to the approved per-path state table;
3. enable amplification or reduce attenuation only when the resulting state satisfies the validated sensitivity and overload criteria;
4. prevent rapid or unstable gain-state switching; and
5. record the antenna, selector position, filter assembly, LNA state, attenuator state, receiver-gain state, and calibration revision with every applicable record.

RF Acceptance and Configuration Control The controlled bill of materials shall identify the antenna-line protection, antenna selector, secondary limiter, DC block, both SP8T selectors, every filter assembly, LNA, bypass device, attenuator, connector, cable, enclosure, and control interface.

Acceptance under **BA-01-RF** shall characterize every approved antenna, filter, gain, attenuation, and bypass combination for insertion loss, ripple, usable bandwidth, stopband rejection, inactive-path isolation, return loss, sensitivity, overload behavior, switching transients, settling time, temperature response, and calibration correction.

Timing and Synchronization

The fixed node shall use a GNSS-disciplined timing subsystem to provide:

- a 10 MHz frequency reference for HackRF receiver-frequency stability; and
- a 1 PPS signal for UTC-referenced host timing and event timestamping.

The timing subsystem shall satisfy the frequency-offset, event-timestamp resolution, maximum event-timestamp error, reference-lock, continuity, startup-readiness, holdover, and loss-of-reference criteria defined in **BA-01-TIM**. The baseline shall not claim UTC-referenced sample timestamps.

A generic userspace GPIO timestamp shall not be used as evidence of compliance with the timestamp-error criterion. Any GPIO-based 1 PPS implementation shall use an approved kernel PPS interface or hardware-assisted

capture method and shall be characterized under the maximum approved processing workload.

The timing service shall record:

- the active timing source and GPSDO identifier;
- GNSS-lock, discipline, and holdover status;
- 10 MHz reference availability;
- 1 PPS continuity and detected discontinuities;
- UTC synchronization status and estimated timestamp error;
- timing calibration and correction identifiers; and
- the validity state associated with each applicable observation.

Following loss of GNSS lock, 10 MHz reference, or 1 PPS continuity, affected records shall be marked with the applicable degraded or invalid timing state. Operational timestamp validity shall not be restored until the reacquisition and stability criteria defined in **BA-01-TIM** have been satisfied.

The HackRF trigger interface may be used for repeatable local laboratory tests or controlled triggered acquisition. It shall not be represented as evidence of UTC sample alignment, phase coherence, inter-node synchronization, or sample-synchronous operation unless those capabilities are separately implemented and validated.

Distributed synchronization, TDOA, phase-coherent processing, and sample-level alignment across nodes remain outside the approved *Template 1S: Fixed Single-Node Monitoring* baseline.

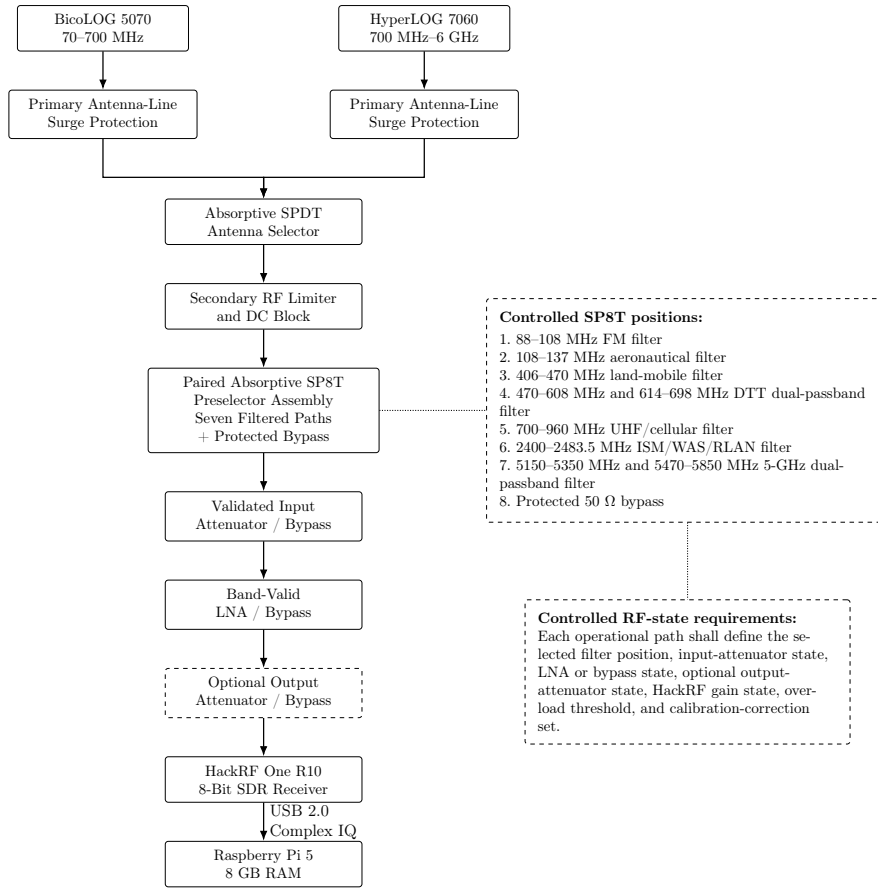


Figure 1: Controlled RF architecture for the fixed-site 70 MHz–6 GHz spectrum-sensing node. Operational monitoring is restricted to the controlled preselector bands and scan plan.

Selected Software Architecture

The software architecture shall separate hardware access, streaming acquisition, signal processing, record management, assurance, security, and network transport. Hardware-specific settings and calibration data shall remain isolated from higher-level detection, record-generation, and export logic.

All deployed software packages, dependencies, configurations, processing parameters, and calibration files shall be identified by version and integrity hash in the controlled software configuration.

Controlled Processing Flow The approved processing sequence shall be:

Layer	Controlled Component	Controlled Responsibility
SDR access	libhackrf and SoapySDR-HackRF	Device discovery, initialization, tuning, sample-rate selection, sample-format control, receiver-gain control, clock-source selection, and isolation of hardware-specific arguments
Streaming acquisition	GNU Radio streaming runtime	Bounded buffering, sample-continuity monitoring, discontinuity detection, DC-offset conditioning, timestamp association, queue-state monitoring, and controlled backpressure or record dropping
Signal processing and detection	GNU Radio, Python, NumPy, and SciPy	FFT and PSD generation, averaging, noise-floor estimation, occupancy calculation, threshold detection, event formation, and approved derived-measurand generation
Record and metadata management	SigMF records and controlled event, health, and audit schemas	Associate each record with the antenna, RF path, gain and attenuation states, receiver settings, timing state, calibration revision, software version, processing parameters, and quality indicators
Local storage and retention	Controlled record manager on approved NVMe storage	Atomic record creation, storage-quota enforcement, retention, overwrite control, congestion detection, recovery, and selected-IQ capture management
Integrity and export assurance	Hash-manifest and audit-record service	Generate integrity hashes, record processing and modification history, authorize exports, and preserve the identity of every exported evidence package
Network transport	ZeroMQ over an approved authenticated and encrypted TCP transport	Transfer only approved metadata, events, health records, derived products, and explicitly authorized IQ records; authentication and encryption shall be provided by the approved transport-security configuration
Service and health management	systemd, watchdog, and health-monitoring services	Controlled startup, dependency ordering, supervision, fault detection, restart limits, degraded-state handling, and shutdown
Host security and	Linux access controls,	Role-based administration,

SDR acquisition → IQ conditioning and continuity monitoring
→ FFT/PSD and derived-measurand generation
→ detection and event formation
→ metadata, health, and integrity records
→ local retention and authorized publication.

Event-triggered IQ recording shall operate as a controlled branch from the event-formation stage. Failure of that branch shall not silently interrupt the generation of continuity, health, or fault records.

Mandatory Runtime Controls The implementation shall:

- use bounded queues and explicitly defined queue-overflow behavior;
- detect and record sample loss, clipping, invalid timestamps, reference-clock loss, processing overruns, storage congestion, and service failures;
- use only the approved FFT window, transform size, overlap, averaging, normalization, and detection-threshold configuration;
- apply detection thresholds to calibrated or consistently normalized PSD estimates;
- associate every output with its source capture, processing configuration, RF-path state, timing state, calibration revision, and software revision;
- prevent raw-IQ publication unless the active controlled configuration explicitly authorizes the transfer;
- generate integrity hashes and auditable records for configuration changes, processing, modification, retention, and export actions;
- preserve fault and quality states across service restarts where required to interpret affected records; and
- enter a controlled degraded or failed state when mandatory acquisition, timing, storage, metadata, or integrity services are unavailable.

Software Acceptance and Change Control Acceptance shall verify simultaneous acquisition, processing, storage, network, timing, and RF-control operation under the maximum approved workload. Testing shall include sample continuity, bounded-queue behavior, service restart, storage congestion, interrupted writes, configuration restoration, record integrity, unauthorized export prevention, and fault recovery.

Alternative software components, processing parameters, metadata schemas, transport mechanisms, or deployment configurations shall require a controlled configuration change, traceability update, and revalidation under the applicable BA-01-ACQ, BA-01-STR, and BA-01-SW, BA-01-SEC, and BA-01-EVD criteria.

Power and Thermal Design

Table 8 provides the provisional power budget used to size the controlled power architecture defined in Table 2. The values are planning estimates and shall be replaced or confirmed using the approved component specifications and measured maximum-load results.

Load	Nominal Planning Load	Maximum Planning Load
Raspberry Pi 5 with active cooling	7.0 W	15.0 W
HackRF One R10	2.0 W	3.0 W
LNA/bypass assembly	0.5 W	0.8 W
Programmable attenuation assembly	0.1 W	0.2 W
Antenna selector, paired SP8T selectors, and control electronics	0.1 W	0.5 W
GPSDO and timing-interface hardware	1.0 W	2.0 W
NVMe storage device and interface	1.5 W	5.0 W
Estimated DC-load subtotal	12.2 W	26.5 W
Conversion and distribution allowance*	1.8 W	4.0 W
Provisional complete-node design load	14.0 W	30.5 W

*The conversion and distribution allowance provisionally represents DC–DC conversion losses, wiring losses, protection-device losses, and unallocated low-power control loads. Acceptance shall use measured values from the final controlled configuration.

Table 8: Provisional complete-node power budget

Power-System Sizing The controlled baseline shall use a regulated complete-node power source rated for at least 40 W of continuous DC output. Relative to the provisional 30.5 W maximum design load, this provides approximately 9.5 W of unallocated DC-output capacity for startup current, load transients, component variation, and controlled configuration changes.

The 40 W rating is a minimum power-source output capability. The MP-01 limit of 50 W applies to measured complete-node input-power consumption under the maximum approved workload, including power-conversion losses.

Each powered branch shall have documented nominal voltage, maximum current, conductor rating, connector rating, protection device, disconnect method, and fault-isolation behavior. The Raspberry Pi 5 shall use its dedicated regulated 5 V/5 A rail; RF, storage, timing, switching, and cooling loads shall use appropriately regulated and independently protected auxiliary branches.

Thermal Design The enclosure and cooling configuration shall be sized for the heat generated at the maximum approved electrical load and the highest ambient temperature defined in BA-01-ENV. Temperature monitoring shall cover, at minimum, the Raspberry Pi 5 processor, NVMe device, HackRF, GPSDO, RF front end, and enclosure air or representative internal thermal reference.

The controlled configuration shall define:

- maximum allowable component and enclosure temperatures;
- temperature-warning and controlled-shutdown thresholds;
- cooling-failure detection and response;
- airflow or conductive heat-transfer requirements;
- startup restrictions following overtemperature shutdown; and
- the temperature and power telemetry retained in health records.

Electrical and Thermal Acceptance Acceptance under BA-01-PWR and BA-01-ENV shall verify:

- steady-state and transient power consumption under the maximum simultaneous acquisition, processing, storage, network, timing, cooling, and RF-control workload;
- startup and inrush-current behavior;
- voltage regulation and ripple at every powered branch;
- branch protection, short-circuit isolation, and recovery;
- controlled recovery following input-power interruption;
- maximum internal temperatures at the approved environmental limits;
- cooling degradation or failure behavior; and
- compliance with the 40 W minimum source capacity and the 50 W complete-node input-power limit.

The final controlled power budget shall use measured maximum values from the accepted hardware configuration. Any change that increases the maximum electrical or thermal load shall require an updated budget and revalidation under the applicable BA-01-PWR and BA-01-ENV criteria.

Cost Baseline

The cost baseline shall distinguish recurring per-node cost, provisional allowances, configuration-dependent items, non-recurring engineering, and five-year lifecycle cost. The USD 1,000–2,000 target defined in MP-01-CST applies only to recurring procurement and node-specific integration cost per operational node.

Current Priced-Item Register The values in Table 9 are preliminary planning values. They are not supplier-validated costs unless supported by a dated quotation in the controlled procurement register.

Table 9: Current preliminary cost register

Item	Qty.	Planning Value (USD)	Cost Classification
HackRF One Pro	1	400	Mandatory; preliminary value
Raspberry Pi 5 with 8 GB RAM	1	175	Mandatory; preliminary value
Raspberry Pi 5 active cooler	1	5	Mandatory; preliminary value
Proxicast ANT-127-002	1	144	Mandatory; converted from supplier price
Diamond D130NJ Super Discone	1	129	Mandatory
Paired SP8T selectors and control hardware	1 set	15	Mandatory; selector set only; part numbers not closed
GPSDO/reference-clock module	1	185	Mandatory; excludes antenna, cables, and PPS interface

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Table 9 – continued

Item	Qty.	Planning Value (USD)	Cost Classification
ZX76-31R75PP-S+ or equivalent	1	222	Mandatory for approved programmable-attenuation paths within its validated range
RF-filter prototype allowance	1	615	Provisional allowance; shall be replaced by separately closed costs for all seven filter assemblies and the bypass implementation
Enclosure and mounting allowance	1	120	Provisional allowance; shall be replaced by the complete accepted enclosure and mounting cost
Cables and connectors allowance	1	50	Provisional allowance; shall be replaced by the production interconnection cost
ZX60-33LN-S+ or equivalent	1	227	Configuration-dependent; included in the planning budget
<i>Equipment subtotal</i>		2,287	Sum of all listed items (mandatory, provisional, and configuration-dependent)

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Table 9 – continued

Item	Qty.	Planning Value (USD)	Cost Classification
<i>Unforeseen expenses (10% contingency)</i>		229	Budget reserve for integration, shipping, fabrication, and procurement uncertainties
Total project budget		2,516	Equipment subtotal (2,287 USD) + contingency (229 USD)

Mandatory Unpriced Configuration Items The controlled procurement bill of materials shall identify, price, and close:

- at least 2 TB of approved NVMe storage and its interface;
- the protected antenna selector and its control hardware;
- primary antenna-line protection for each antenna feed;
- the secondary limiter and DC block;
- all seven filter assemblies, the protected bypass path, internal combining networks, shielding, connectors, and assembly hardware;
- any selected VHF, lower-UHF, general-purpose, or 5-GHz LNA;
- any selected 5-GHz attenuator;
- the GPSDO antenna, timing cables, PPS interface, and mounting hardware;
- the complete regulated power source, conversion modules, distribution hardware, branch protection, wiring, and disconnects;
- the final fixed-site enclosure, mounting plate, cooling, temperature sensors, protected entries, grounding, bonding, and surge-protection hardware;
- production RF cables, adapters, connectors, Ethernet hardware, and mechanical supports;
- shipping, importation, taxes, procurement fees, and supplier minimum-order effects;

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- node-specific assembly, installation, and commissioning; and
 - initial operational spares and the approved contingency.

A provisional allowance shall be removed when the corresponding controlled item is priced. The closed cost shall replace the allowance and shall not be added to it.

Recurring Per-Node Cost The recurring cost of one accepted operational node shall be calculated as:

$$C_{\text{node}} = C_{\text{hardware}} + C_{\text{conditional}} + C_{\text{integration}} + C_{\text{procurement}} + C_{\text{spares}} + C_{\text{contingency}}.$$

The controlled cost register shall define the quantity, supplier, quotation date, currency basis, exchange-rate basis where applicable, and inclusion or exclusion of taxes for every cost entry.

Non-Recurring and First-Article Costs The non-recurring cost register shall separately record RF-filter and enclosure design, prototype fabrication, corrective redesign, calibration-method development, software integration, cybersecurity hardening, acceptance-procedure development, laboratory fixtures, reference equipment, qualification testing, and controlled documentation.

The first-article cost shall be reported as:

$$C_{\text{first}} = C_{\text{node}} + C_{\text{NRE}}.$$

If non-recurring costs are distributed across multiple nodes, the allocation method and production quantity shall be documented.

Cost-Control Decision Budget compliance shall be evaluated only after every mandatory item has a controlled quantity and cost, every provisional allowance has been replaced, every selected conditional item has been included, and procurement, integration, spares, and contingency have been closed.

If $C_{\text{node}} > 2,000$ USD, the project shall approve a budget exception, revise only optional capabilities without violating MP-01 or BA-01, or reopen the platform-selection decision.

Mandatory RF protection, timing, storage, power, enclosure, thermal, grounding, calibration, cybersecurity, sensing-performance, or acceptance requirements shall not be removed solely to satisfy the target.

Lifecycle Cost The five-year lifecycle budget shall be maintained separately from C_{node} and C_{first} . It shall include scheduled calibration, replacement storage, component failures, spare units, software maintenance, field support, site maintenance, backup resources, and component-obsolescence mitigation.

Key Limitations and Controls

Table 10 identifies residual platform limitations that remain after implementation of the selected controls. A mitigation shall not be treated as closure until the applicable BA-01 criterion has been verified and the evidence has been registered in VTM-01.

Table 10: Residual platform limitations and mandatory controls

Residual Limitation	Operational Consequence	Mandatory Control and Acceptance Anchor
8-bit quantization and limited instantaneous dynamic range	Strong signals may cause overload, clipping, desensitization, or masking of weaker signals	Preselection, validated gain and attenuation states, overload detection, and dynamic-range testing under BA-01-RF and BA-01-MET
Single-tuner sequential scanning	Separated bands cannot be observed simultaneously; scan gaps may miss short events	Controlled dwell, revisit, scan-cycle, event-priority, interruption, recovery, and missed-event criteria under SP-01, BA-01-ACQ, and BA-01-MET
USB 2.0 acquisition interface	Sustained-transfer margin is limited at higher sample rates and simultaneous host workloads	Maintain the validated continuous mode and verify sample continuity, queue behavior, and alternate modes under BA-01-ACQ
RF-path-dependent sensitivity and insertion loss	Detection performance and calibration vary by antenna, filter, LNA, attenuation, and bypass state	Characterize every approved RF-path combination under BA-01-RF and verify sensing outputs under BA-01-MET
Conditional active-component coverage	General-purpose gain and attenuation components do not cover every approved band	Define an accepted LNA/bypass and attenuator/bypass state for every filter path under BA-01-RF

Continued on next page

Table 10 – continued

Residual Limitation	Operational Consequence	Mandatory Control and Acceptance Anchor
Unclosed composite filter assemblies	DTT and 5-GHz path feasibility, loss, rejection, integration, and cost remain uncertain	Complete prototype, RF, environmental, integration, and cost closure before configuration freeze
Host, storage, and processing contention	Simultaneous DSP, recording, networking, and storage may cause queue overflow, incomplete records, latency, or sample loss	Bounded queues, explicit overflow behavior, storage-reserve controls, health monitoring, and maximum-workload testing under BA-01-ACQ, BA-01-STR, and BA-01-SW
Host-assigned event timing	Valid UTC event timestamps do not establish UTC-aligned sample indices, phase coherence, or inter-node synchronization	Verify timestamp origin, resolution, maximum error, uncertainty, holdover, and continuity under BA-01-TIM
Single-node observation	The baseline cannot independently provide TDOA, direction finding, geolocation, or cross-node confirmation	Restrict outputs to the approved fixed single-node scope; require a separate profile and acceptance specification for distributed capabilities
Environmental and thermal dependence	Temperature, humidity, cooling degradation, and enclosure conditions may affect stability, storage, timing, and calibration	Verify maximum-load operation and environmental limits under BA-01-PWR, BA-01-ENV, and BA-01-SIT
Screening-grade evidentiary status	Node records alone are insufficient for an independent regulatory non-compliance finding	Preserve calibration, provenance, timing, quality, and integrity metadata and apply Template 5 controls under BA-01-EVD when formally invoked

Any residual limitation accepted without full mitigation shall be recorded as an explicit restriction or condition in EAM-01.

Validation and Deployment Conditions

Preliminary laboratory configurations may generate controlled test and calibration evidence but shall not generate operational monitoring records or constitute platform acceptance.

Operational release of a fixed-site node shall require:

1. closure and configuration control of the complete hardware, software, security, installation, and cost baselines;
2. approval of the applicable MP-01, SP-01, and BA-01 revisions;
3. laboratory qualification under Q3-01;
4. successful execution of the applicable T0-* procedures for BA-01-MET, BA-01-ACQ, BA-01-RF, BA-01-TIM, BA-01-STR, BA-01-PWR, BA-01-ENV, BA-01-SW, BA-01-SEC, BA-01-SIT, and BA-01-EVD;
5. complete requirement-to-evidence traceability in VTM-01;
6. valid calibration, configuration, software-version, security, timing, site, and integrity records for the released node; and
7. an approved EAM-01 decision authorizing operational use.

An *Accepted* decision shall authorize operation only within the configuration, site, monitoring scope, scan plan, acquisition modes, security state, and environmental conditions represented by the supporting evidence.

A *Conditionally Accepted* decision shall authorize restricted operation only when EAM-01 identifies the unresolved conditions, permitted restrictions, affected requirements, corrective actions, verification evidence, responsible authority, and closure or expiration criterion.

A *Rejected* decision shall prohibit operational deployment.

Any change to the accepted hardware, software, RF path, timing subsystem, power architecture, enclosure, site installation, scan plan, calibration, security configuration, or sensing algorithm shall undergo documented impact assessment. The affected tests and traceability links shall be repeated or updated before the modified configuration returns to service.

Operational records shall identify the accepted configuration revision, SP-01 revision, calibration state, software revision, security configuration revision, site identifier, timing-validity state, and applicable EAM-01 decision.

Summary

The HackRF One R10 and Raspberry Pi 5 are conditionally selected for the fixed-site, single-node screening scope defined in MP-01. Operational release remains blocked pending closure of the sensing-performance criteria, scan-coverage requirements, composite RF-filter assemblies, attenuation policy, secure transport configuration, software and cybersecurity acceptance, site commissioning, controlled procurement cost, and the applicable EAM-01 decision.

Inherent residual limitations include 8-bit dynamic range, sequential scan gaps, RF-path-dependent sensitivity, USB transfer margin, host-assigned event timing, and the absence of distributed or localization capability.

A Regulatory Band Context and Controlled Sources

Table 11 is the authoritative band list adopted by MP-01. Exact operational channels, exclusions, priorities, and effective assignments shall be controlled through SP-01.

For every row, the MP-01 regulatory-source register shall identify the issuing authority, source title, document identifier, revision, effective date, retrieval date, geographic scope, applicability, and freshness status. The table shall not be interpreted as evidence of ANE authorization, delegation, endorsement, or operational responsibility.

Frequency Band	Controlled Monitoring Scope
88–108 MHz	FM sound broadcasting; technical verification and adjacent-band interference screening
108–137 MHz	Aeronautical radionavigation and communications; safety-critical interference screening
406–470 MHz	Private land-mobile services
470–608 MHz and 614–698 MHz	DTT/TDT using DVB-T2, corresponding to UHF channels 14–36 and 38–51; 608–614 MHz remains excluded
700–960 MHz	Approved 700-, 800-, and 900-MHz mobile-service monitoring assignments
2400–2483.5 MHz	ISM and wireless-access systems, including Wi-Fi and Bluetooth
5150–5250 MHz, 5250–5350 MHz, 5470–5725 MHz, and 5725–5850 MHz	Approved 5-GHz WAS/RLAN subbands; 5350–5470 MHz remains excluded

Table 11: Regulatory band context adopted by MP-01